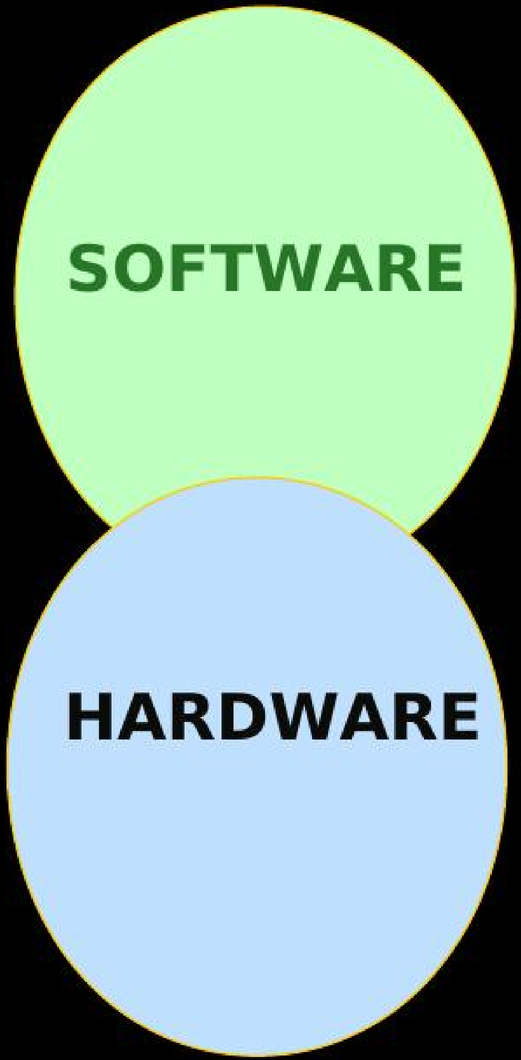
Chapter1 Introduction to Computers

What are computers?

Computers are electronic devices that can follow instructions **to** accept input, process the input and then produce **information**.

Look inside the computer



Computers are made of

1. HARDWARE
2. SOFTWARE

Hardware

غير مطلوبة فقط للفهم

Hardware

The parts of computer itself (tangible objects) including :

- CPU (or Processor) and Primary memory (or Main Memory)
- Input devices i.e the keyboard and mouse
- Output devices
- Storage devices

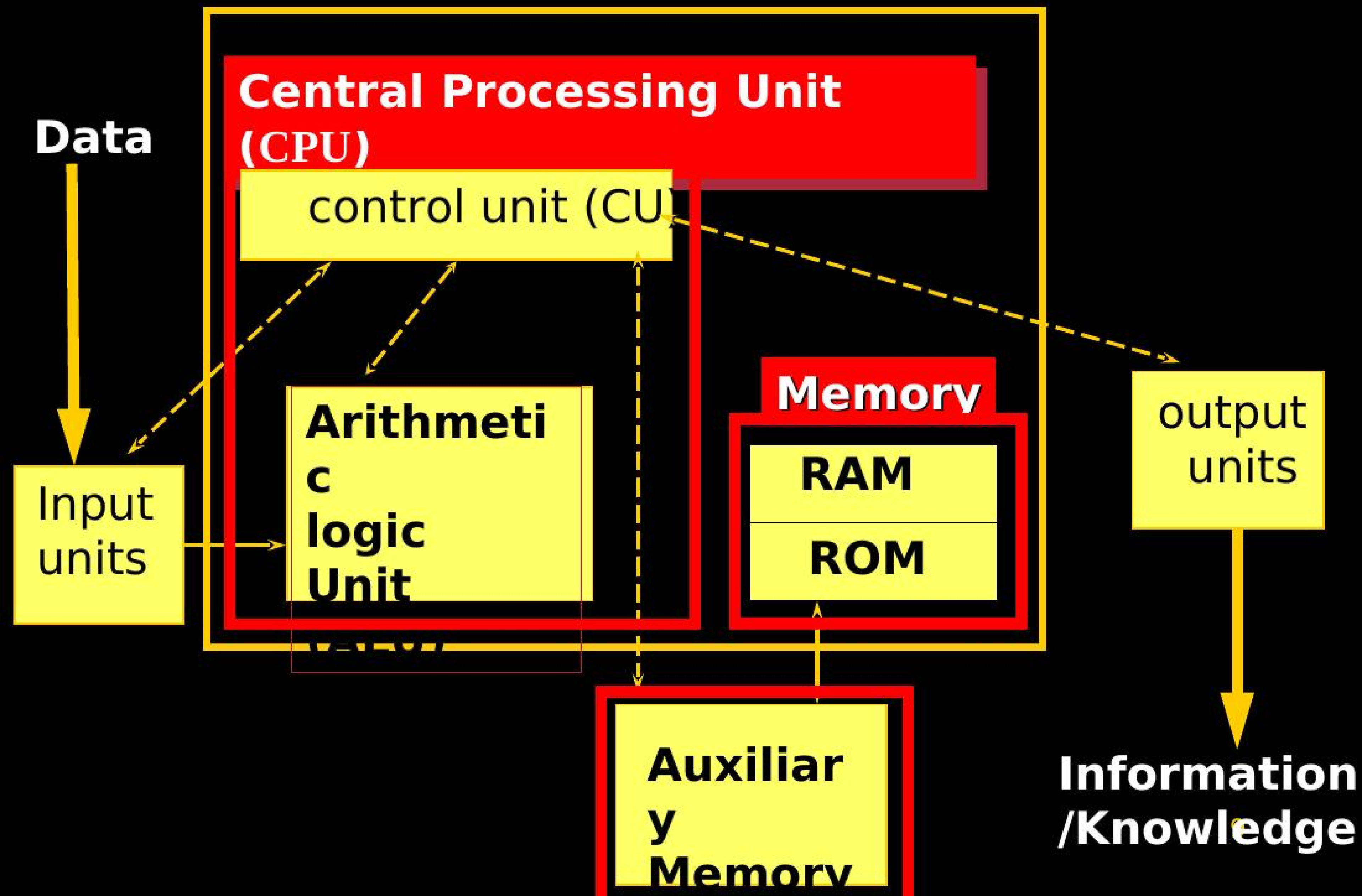
The Case (System Unit or System Cabinet)

غير مطلوبة فقط للفهم

Hardware

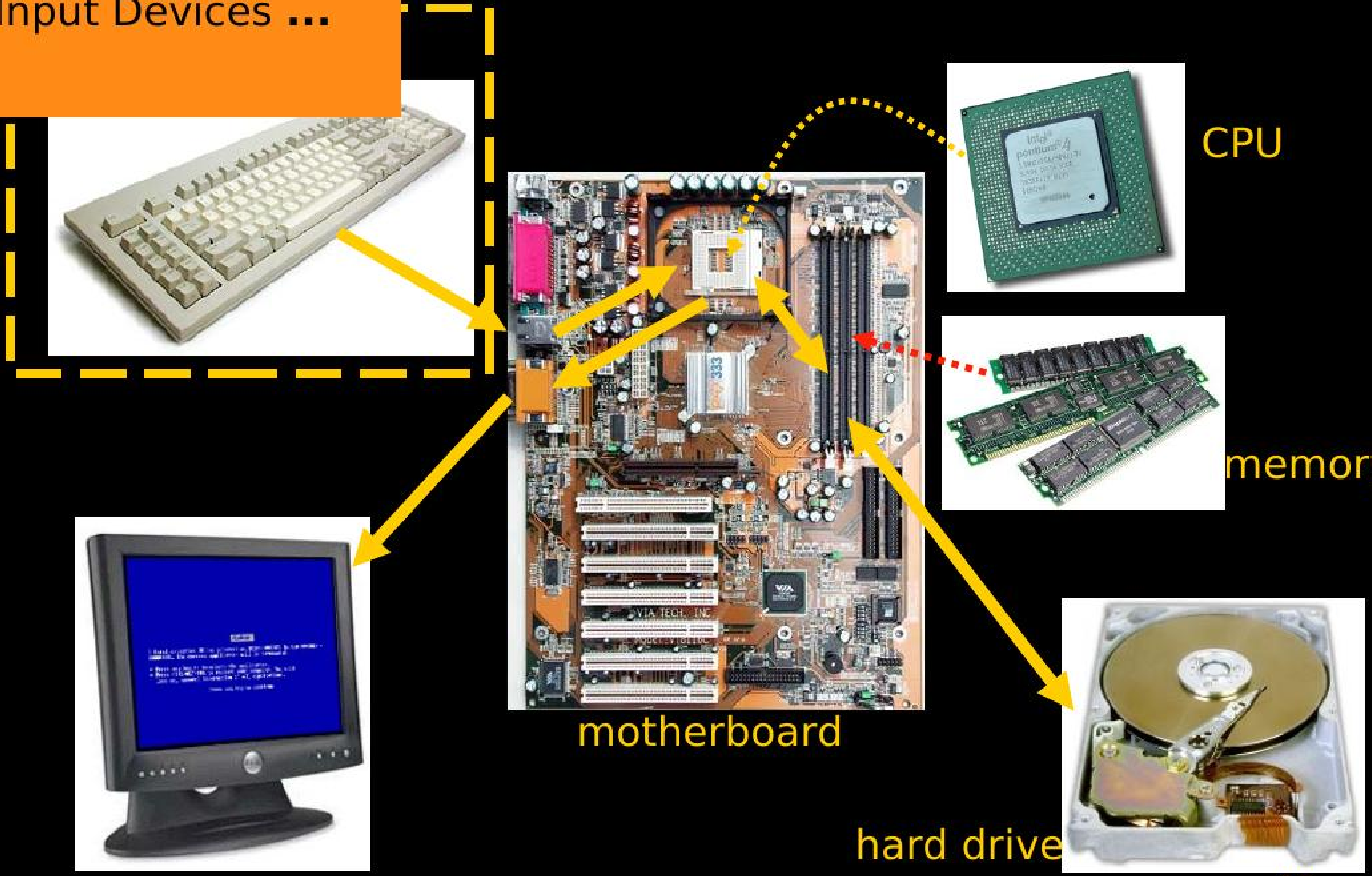
1. Central Processing Unit (CPU)
2. Input units
3. Output units
4. Memory (Main or Primary Memory & Secondary or Auxiliary Memory)

omponents of a Computer Syste



Hardware Organization

Input Devices **...**



غير مطلوبة فقط للفهم

Input Devices

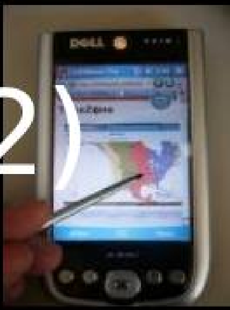
- Translate data from **form** that humans understand to one that the computer can work with
- Most common are keyboard and mouse

Examples of Input Devices

1. Keyboard (QWERTY keyboard, ATMs keyboard)
   ATM: automatic teller machine
2. Mouse
3. Scanner
4. Pre-storage Devise (Disk, CD's, … etc.)
5. Optical mark recognition (Light Pin , Bar code scanners)
6. Microphone
7. Joystick .

See Page 4 in text book

Examples of Input Devices(2)



8. Point and Draw devices
9. Trackball
10. Touchpad
11. Touch screen
12. Magnetic stripes and smart cars.
13. Digital Cameras

Trackball

Trackpoint

Digitizer Tablet
and Pen

Photos Courtesy of:
• Imation Corporation
•Houston Instrument Div. AMETEK, Inc.
•Advanced Gravis Computer Technology Ltd.

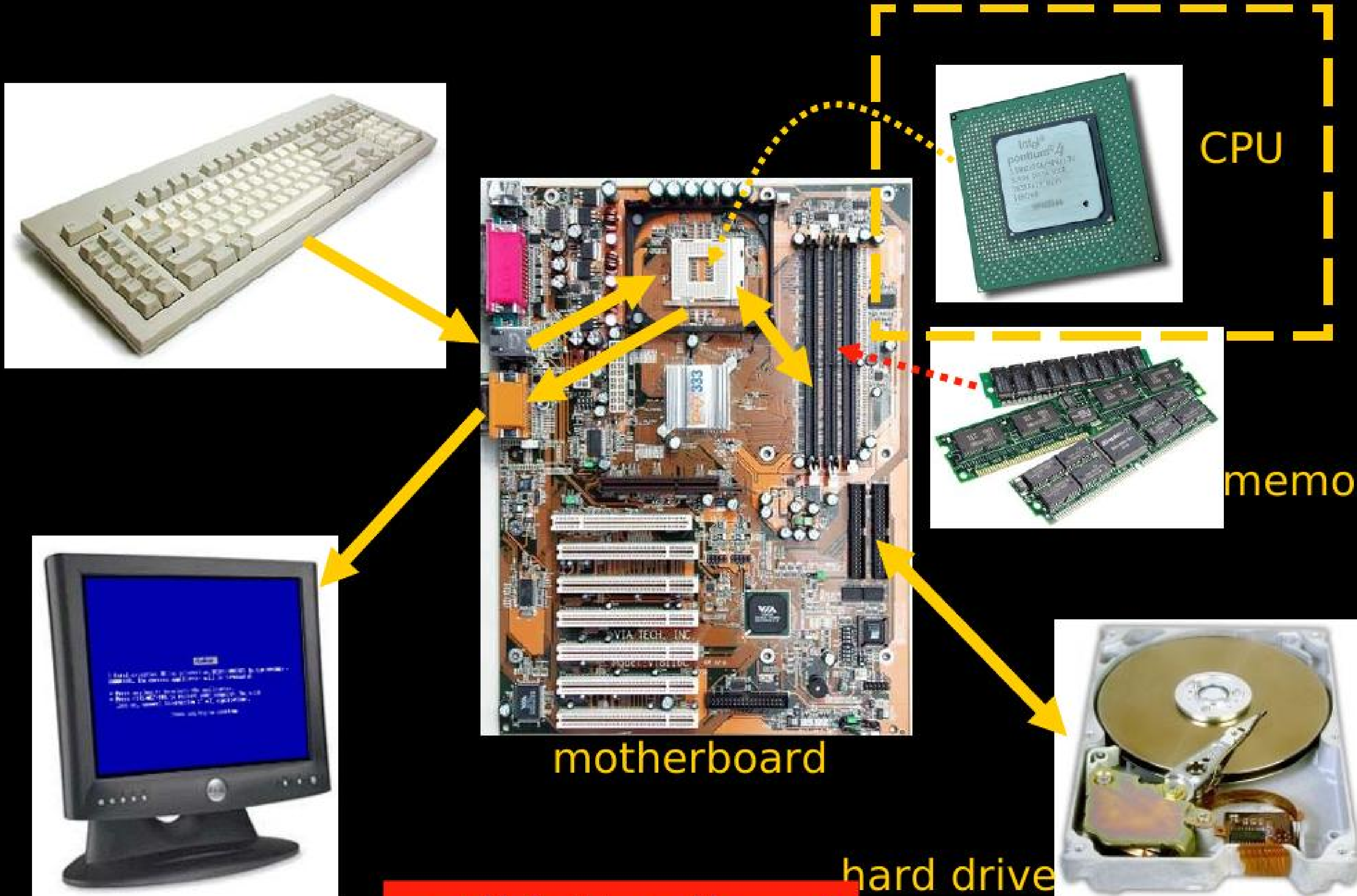
Joystick

Trackpad

Hardware Organization

غير مطلوبة فقط للفهم

Hardware Organization

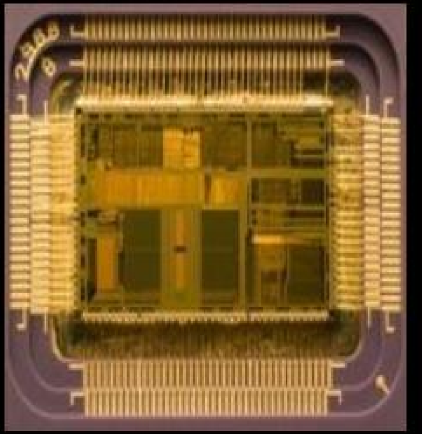


غير مطلوبة فقط للفهم

Central Processing Unit (CPU)

- A specific chip or the processor

  a CPU's performance is determined by the rest of the computers circuitry and chips.
- The Central Processing Unit (CPU) performs the actual processing of data
- The speed (clock speed) of CPU measured by Hertz (MHz)



The CPU consists of :

- Control Unit (CU)
- Arithmetic and Logical Unit (ALU)
- Some Registers

The Control Unit (CU) :

coordinates all activities of the computer by:

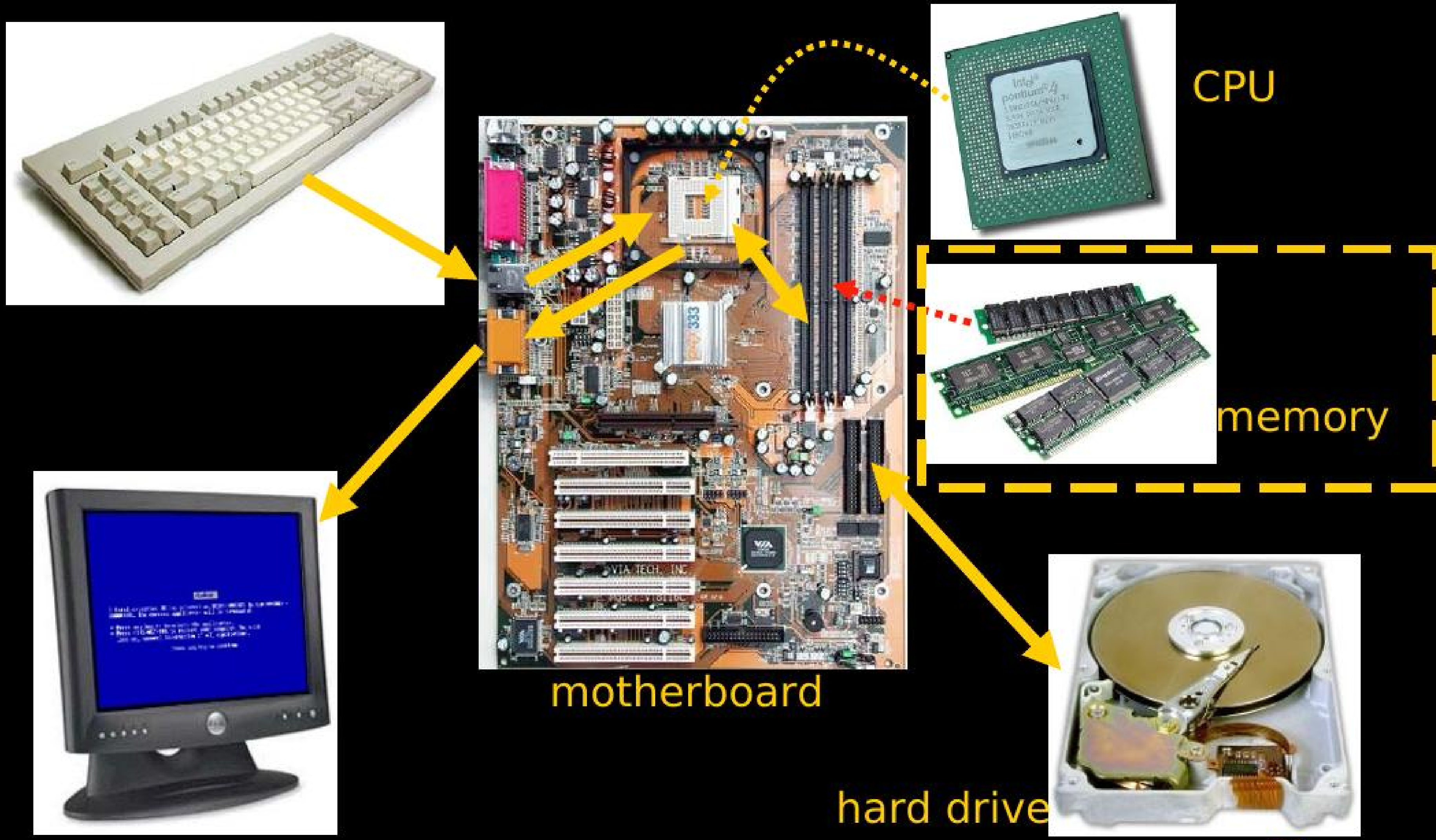
- Determining which operations to perform and in what order to carry them out.
- The CU transmits coordinating control signals to other computer components.

The ALU :

consists of electronic circuitry to perform:

- Arithmetic operations (addition, subtraction, multiplication and division)
- Logical operations (and, or, not, ...) and to make some comparisons (less-than, equal, ... etc.)

Hardware Organization



غير مطلوبة فقط للفهم

Primary Memory

- Memory (fast, expensive, short-term memory): Enables a computer to store, at least temporarily, data, programs, and intermediate results.

- Two general parts:

  1. RAM
  2. ROM

RAM (Main Memory)

- its a primary storage or random access memory (RAM).
- it temporarily holds data and programs for use during processing (volatile)
- Any information stored in RAM is lost when the computer is turned off.
- RAM is the memory that the computer uses to temporarily store the information as it is being processed. The more information being processed the more RAM the computer needs.
- RAM consists of locations or cells. Each cell has a unique address which distinguishes it from other cells.

ROM: Read Only Memory

ROM is part of memory

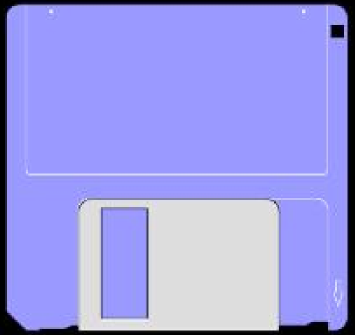
- Programmed at manufacturing time
- Its contents cannot be changed by users
- It is a permanent store

Secondary Storage

- Stores data and programs permanently: its retained after the power is turned o

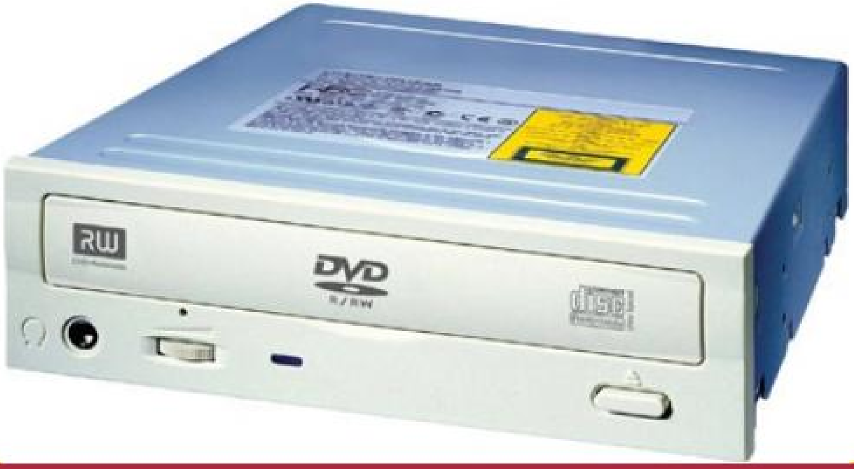
- Examples
  - Hard Drive (Hard Disk)

    Located outside the CPU, but most often contained in the system cabinet



  - Floppy Disk
  - Optical Laser Discs
    - CD-ROM, CD-RW, and DVD

Kinds of Disk **Drives**



Common Secondary Media

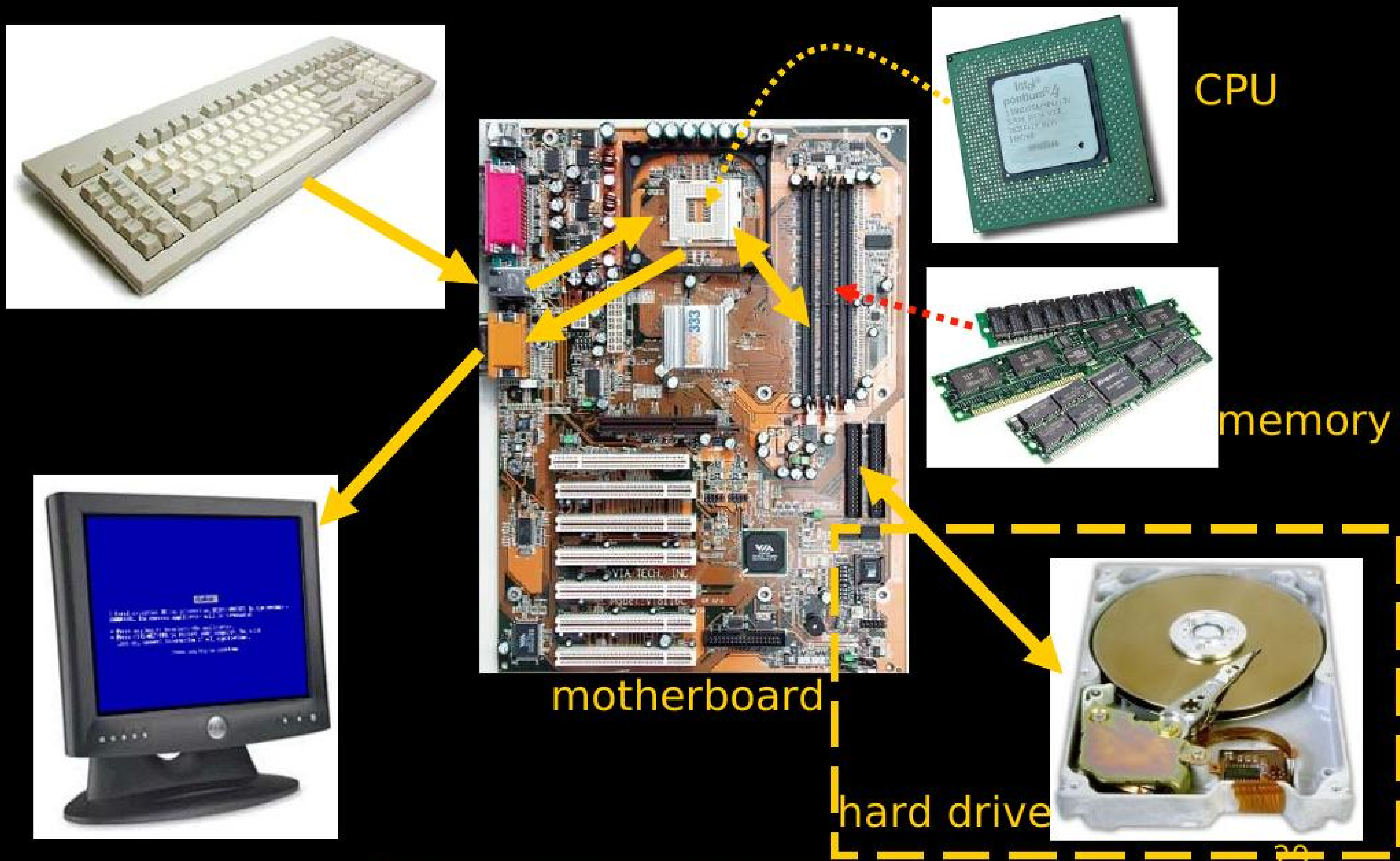
- **Diskettes**
  - Data represented as magnetic spots on removable flexible plastic disks
  - Most common size is 3 1/2 inches, in a rigid plastic case
  - Disk drive holds the diskette, reads or retrieves the data and writes or stores data

Common Secondary Media

- Hard drive

  - Data is represented magnetically as with diskettes
  - Normally more than one rigid platter in a sealed unit
  - These disks are not removable
  - Significantly more capacity and faster operating than diskettes

Hardware Organization



غير مطلوبة فقط للفهم

Common Secondary Media

- Optical Laser Discs
  - CD ROM & DVD's
    - Data is represented as pits and lands
    - Some kinds are read only (CD-ROM) and some Kinds are rewritable (CD-RW)
    - Significantly more capacity and faster operating than diskettes

DVD: Digital Video Disk

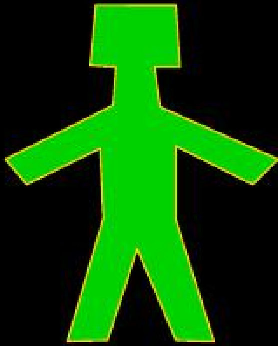
Hardware Organization

غير مطلوبة فقط للفهم

Output Devices

Pieces of equipment that translate the processed information from the CPU into a form that humans can understand.

Processed information



Output Devices

- Monitors
- Printers
  - Dot matrix printers
  - Ink jet printers
  - Laser printers
- Sound Blasters (Sound Card By Creative Lab)
- Controlling other devices

Software

The instructions that tell the computer what to do

1. Application Software - helps end-users perform general purpose tasks
2. System Software - enables application software to interact with the computer

System Software

The most important

System Software

is the

Operating System

:Examples of operating systems
Windows XP, DOS, Apple, UNIX

System Software

- The software that controls everything that happens in a computer.
- Background software, manages the computer's internal resources

Resources examples : CPU, RAM , I/O devices, …

Application Software – Basic Tools

- Word processors– **example: Microsoft word**
- Spreadsheets-- **example: Microsoft Excel**
- Database managers-- **example: Microsoft Access**
- Graphics-- **example**: **Photoshop**

Spreadsheets: Computer software that allows the user to enter columns and rows of numbers in a accounting book like

Units of Measurements

- Bit (Binary Digit)(takes two values: 1 or 0)
- Byte = 8 bits

- KB (Kilo-byte) = 1024 bytes
- MB (mega-byte) = 1024 KB
- GB (giga-byte) = 1024 MB
- TB (Tera-byte) = 1024 GB

Remark: $1024=2^{10}$

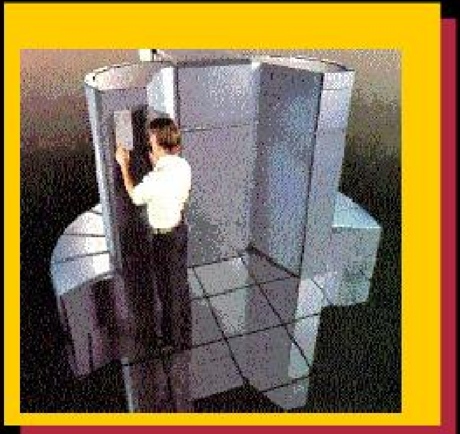
Four Kinds of Computers

1. Microcomputers

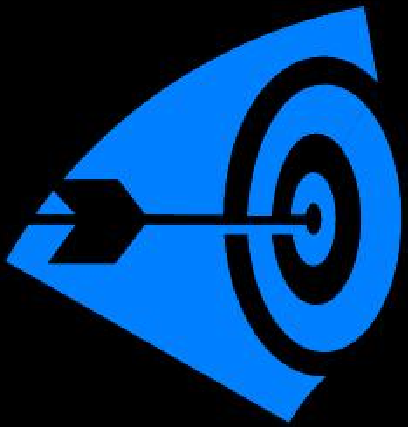
2. Minicomputers

3. Mainframe computers

4. Supercomputers



- Microcomputer =>Personal Computer => PC
- There are **3** types of the Microcomputers :
  1. Laptop
  2. Desktop
  3. Workstation



- Personal Computer: A small, single-user computer based on a microprocessor.
- Workstation: A powerful, single-user computer. A workstation is like a personal computer, but it has :
  - a more powerful microprocessor and,
  - in general, a higher-quality monitor.

Minicomputer, Mainframe, and Supercomputer

- Minicomputer: A multi-user computer capable of supporting up to hundreds of users **simultaneously**.
- Mainframe: A powerful multi-user computer capable of supporting many hundreds or thousands of users **simultaneously**.
- Supercomputer: An extremely fast computer that can perform hundreds of millions of instructions per second.

Minicomputers

- Desk-sized
- More processing speed and storage capacity than microcomputers
- General data processing needs at small companies
- Larger companies use them for specific purposes

Mainframe Computers

- Larger machines with special wiring and environmental controls
- Faster processing and greater storage than minicomputers
- Typical machine in large organizations

Supercomputers

- The most powerful of the four categories
- Used by very large organizations, particularly for very math-intensive types of tasks

Supercomputers

haracteristics of Computers

1- Store a large amount of data an
information for a long period of time.

2- process data and information in high accuracy level .

3- Speed in processing data information.

Understanding the difference between Data, Information and Knowledge:

- ✓ <u>**Data**</u>: is the name given to basic facts such as names and numbers.
- ✓ <u>**Information**</u>: is data that has been converted into a more useful or intelligible form.
- ✓ <u>**Knowledge**</u>: arrangement of information and classifying information of the same type or the same topic.

- e. g.

-2 4 0 -3 10 (**data**)

|| sort

\ /

-3 -2 0 4 10

(**information**)

Processing **data** produces **information**, and processing information produces **knowledge**.

Computer Viruses

Computer Viruses

- A computer virus is an application program designed and written to destroy other programs.

- It has the ability to:
  - Link itself to other programs
  - Copy itself (it looks as if it repeats itself)

Examples of Viruses

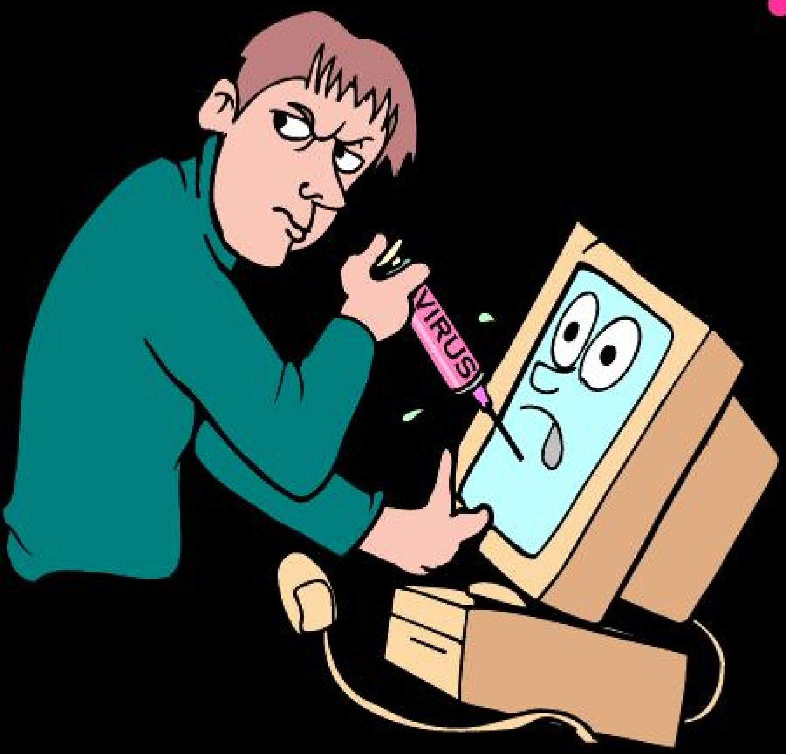
- Monkes
- ABC
- Crabs
- CIH

Viruses and Virus Protection

- A virus program
  - Infects programs, documents, databases and more …
  - It is man-made
  - It can hide and reproduce
  - It can lay dormant (inactive) and then activate

Anti-virus programs can help

Sources of Computer Viruses



- Three primary sources
  - The Internet
    - Via downloads and exchanges
  - Diskettes
    - Exchanging disks
  - Computer networks
    - Can spread from one network to another

How do you know if you have a virus?

- Lack of storage capability
- Decrease in the speed of executing programs
- Unexpected error messages
- Halting the system

Virus Protection

- The software package distributed with new PCs always includes an **antiviral program.** The best way to cope with viruses is to recognize their existence and use an antiviral, or antivirus program.